Project 1: Fluid System Transient Analysis

Alex Knezevic, Arthur Kot, Aidan O'Sullivan, Athiwat Chotrapeepirom

10/30/2026

1 The Premise

The premise of this project is to model a transient response of a fluid system found in collegiate liquid rocketry. This fluid system is modeled after Liquid Rocketry at Illinois's (LRI) Propulsion Module design, and takes real constraints from that project since one of our members (A. Knezevic) was a part of its design. The objective of the fluid system modeled is to get fuel (70% isopropyl alcohol, IPA) to an engine at a constant mass flow via a pressurant (nitrogen). The structure of this system is detailed in Figure 1 below; this is the system we will model, with constraints given by the engine in bold and design choice or calculated constraints in italics.

The system consists of a pressurant tank filled with nitrogen gas, and a regulator regulating the high pressurant tank pressure down to a desired ullage pressure (ullage: space filled by gas in the fuel tank). The gas then runs through plumbing with a certain C_dA (math will be detailed later) to the ullage of the fuel tank. This pressurizes the fuel, which flows through plumbing with its own C_dA to the engine, which is at 300 psi.

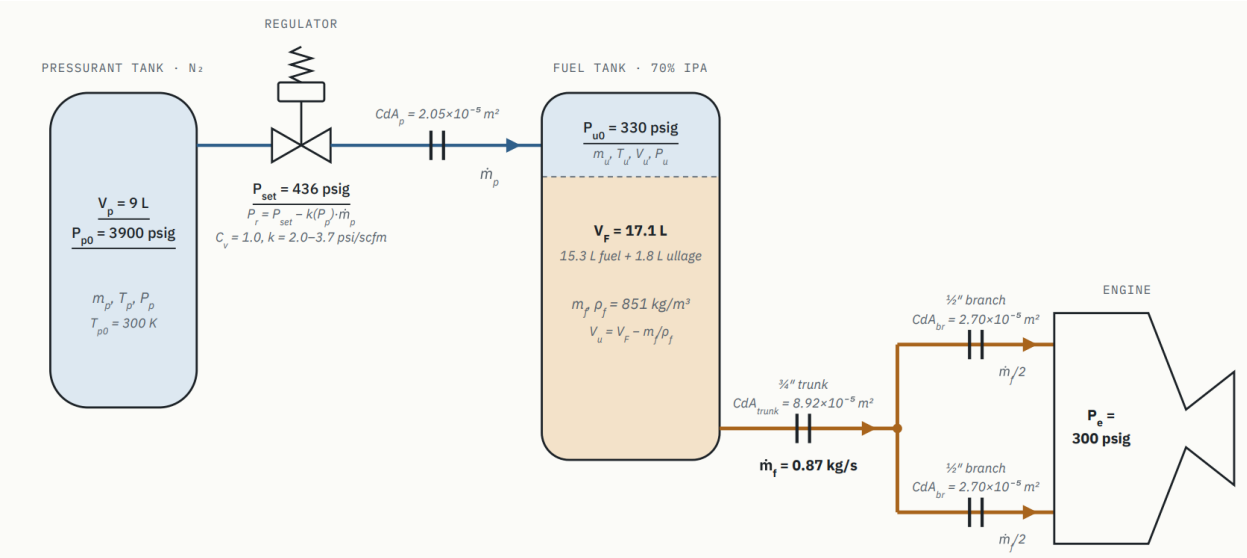


Figure 1: Feed system schematic. Bold: requirements set by the engine. Underlined: design inputs. Italic: design choice or calculated constraints.

The underlined constraints are the main design choices that this model can help to optimize. They are the main "inputs" to the system model, basically the values to play around with. A large part of the use case of modeling this system is as a design tool to try out different parameters of possible configurations and see how the system reacts. Having multiple test cases of different configurations helps justify component choice, especially when it comes to the most costly component: pressure regulators.

1.1 Regulator model

The response of this system is largely influenced by the choice of regulator, since it is the flow rate bottleneck in this system. We are modeling a spring regulator controlling the flow rate of the system, whose flow rate is dependent on the pressure difference between the set pressure, what the regulator will equilibrate at, and the actual pressure after the regulator. This relation is roughly linear and we approximate it as such:

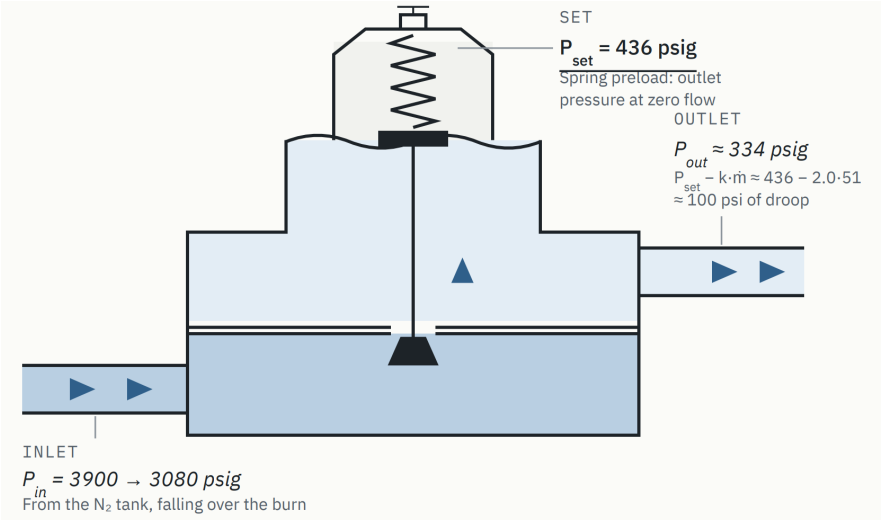


Figure 2: Regulator outlet pressure versus nitrogen flow (Swagelok KPF, $C_v = 1.0$).

$$P_{out} = P_{set} - K\,\dot{V}, \qquad \dot{V} = \frac{\dot{m}_p}{\rho_{std}} \quad \Longrightarrow \quad \dot{m}_p = \rho_{std}\,\frac{P_{set} - P_{out}}{K} \tag{1}$$

where $\dot{V}$ is the volumetric flow rate at standard conditions and $\rho_{std} = P_{std}/(R\,T_{std})$ is the nitrogen density at 60 °F and 14.696 psia. For nitrogen, $\dot{V}\,[\text{scfm}] \approx 1792\,\dot{m}_p\,[\text{kg/s}]$.

K is the droop of the regulator in psi/scfm, which represents how good the regulator is at regulating the pressure of the system. A higher flow rate (lower K) at a given difference between set and outlet pressure gives a faster response and tighter regulation in a flowing system. Droop is only a function of inlet pressure and temperature in this context (as flow is always choked), so it gets "worse" at regulating pressure as the flow goes on and inlet pressure decreases.

In the model, K is taken from the manufacturer's curves at two inlet pressures and interpolated linearly in inlet pressure, held constant outside that range:

$$K(P_{in}) = K_1 + (K_2 - K_1)\,\frac{P_{in} - P_1}{P_2 - P_1}, \qquad \begin{aligned} P_1 &= 1000 \text{ psig}, \quad K_1 = 3.7 \text{ psi/scfm} \\ P_2 &= 3000 \text{ psig}, \quad K_2 = 2.0 \text{ psi/scfm} \end{aligned} \tag{2}$$

These two points scale as $K \propto P_{in}^{-0.56}$.

1.2 Plumbing and the discharge coefficient

How this factors into the flow rates beyond the regulator, gas and fuel, is based on the discharge coefficient equation. We use the form of this equation that gives us C_dA on one side, which is constant for the section of plumbing it describes. You can think of C_dA as similar to the "friction factor" that the fluid experiences: given a ΔP across a section, the C_dA dictates the flow rate. A straw has a smaller C_dA than a hose and is harder to blow through to get the same flow rate.

We used actual values for C_dA from the design choices LRI went with for the Propulsion Module (Table 1). These C_dAs gave us the equations to relate flow rates with pressure, which is the fundamental basis of how we modeled this system.

Incompressible (fuel lines, and gas lines with $\Delta P/P \lesssim 5$–10%):

$$C_dA = \frac{\dot{m}}{\sqrt{2\rho\,\Delta P}}, \qquad \text{gas: } \rho = \frac{P}{RT} \implies C_dA = \dot{m}\sqrt{\frac{RT}{2P\,\Delta P}} \tag{3}$$

Combining sections. Elements in series and in parallel combine as

$$C_dA_{series} = \left[\sum_i \frac{1}{(C_dA)_i^2}\right]^{-1/2}, \qquad C_dA_{parallel} = \sum_i (C_dA)_i, \tag{4}$$

The fuel feed is one $\frac{3}{4}''$ trunk in series with two parallel $\frac{1}{2}''$ branches:

$$C_dA_F = \left[\frac{1}{C_dA_{trunk}^2} + \frac{1}{(2\,C_dA_{br})^2}\right]^{-1/2} \tag{5}$$

Table 1: Plumbing C_dA values and the regime of each element during the burn.

Element	C_dA (m^2)	Regime	Form used
Pressurant line	2.05×10^{-5}	Subsonic, $\Delta P/P \approx 1\%$	Eq. (3)
$\frac{3}{4}''$ fuel trunk	8.917×10^{-5}	Liquid	Eq. (3)
$\frac{1}{2}''$ fuel branch (each, 2 in parallel)	2.700×10^{-5}	Liquid	Eq. (3)
Total fuel feed	4.619×10^{-5}	Liquid	Eq. (5)

1.3 Formulation as an initial value problem

Since this regulator does not have a tight enough regulation to be steady state, we have to model it as an initial value problem (IVP) and see how the system responds over the course of the flow. With the limited pressurant capacity as well, the flow never reaches a full steady state, and cannot be approximated as such. The regulator's K value changes significantly through the course of the flow, where steady-state approximations are not adequate for describing how the regulator responds. Along with this, the nonlinearity of the pressure–flow rate equations makes this an IVP that can only be modeled with numerical methods, and cannot be solved analytically.

To model this system mathematically, we need the equations given previously and other elementary thermodynamic equations (ideal gas law, mass conservation, etc.) to put in the form:

$$\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}), \qquad \mathbf{y} = \begin{bmatrix} m_p & T_p & m_u & T_u & m_f \end{bmatrix}^T \tag{6}$$

where m_p, T_p are the mass and temperature of nitrogen in the pressurant tank, m_u, T_u are those of the ullage gas, and m_f is the fuel mass remaining. Once we have the system organized like this, we can apply methods and analyze the behavior of this system like any other IVP. It makes sense to organize the system in this way because the state (mass, temperature $\rightarrow$ pressure) affects derivatives of the state (mass flow rate).

State equations. From mass conservation on each volume, isentropic expansion of the pressurant tank, and the first law applied to the ullage:

$$\frac{dm_p}{dt} = -\dot{m}_p \tag{7a}$$

$$\frac{dT_p}{dt} = -(\gamma - 1)\,\frac{T_p\,\dot{m}_p}{m_p} \tag{7b}$$

$$\frac{dm_u}{dt} = \dot{m}_p \tag{7c}$$

$$\frac{dT_u}{dt} = \frac{\dot{m}_p\,(\gamma T_p - T_u) - P_u\,\dot{m}_f/(\rho_f c_v)}{m_u} \tag{7d}$$

$$\frac{dm_f}{dt} = -\dot{m}_f \tag{7e}$$

Algebraic relations. At each evaluation of $\mathbf{f}(\mathbf{y})$:

$$P_p = \frac{m_p R T_p}{\underline{V_p}}, \qquad V_u = \mathbf{V_F} - \frac{m_f}{\rho_f}, \qquad P_u = \frac{m_u R T_u}{V_u} \tag{8a}$$

$$\dot{m}_f = C_d A_F \sqrt{2\rho_f\,(P_u - \mathbf{P_e})} \tag{8b}$$

The nitrogen flow satisfies the regulator droop law, Eq. (1), and the pressurant line loss, Eq. (3), simultaneously:

$$\underline{P_{set}} - K'\dot{m}_p - \frac{\dot{m}_p^2 R T_p}{2P_u\,C_d A_p^2} = P_u, \qquad K' = \frac{K(P_p)}{\rho_{std}} \tag{9}$$

which is quadratic in $\dot{m}_p$:

$$\dot{m}_p = \frac{-K' + \sqrt{K'^2 + 4a\,\left(\underline{P_{set}} - P_u\right)}}{2a}, \qquad a = \frac{R T_p}{2P_u\,C_d A_p^2} \tag{10}$$

with $\dot{m}_p = 0$ when $P_u \geq \underline{P_{set}}$.

Initial conditions and termination.

$$m_p(0) = \frac{P_{p0}\,\underline{V_p}}{R\,T_{p0}}, \quad T_p(0) = T_{p0}, \quad m_u(0) = \frac{\underline{P_{u0}}\,V_{u0}}{R\,T_{u0}}, \quad T_u(0) = T_{u0}, \quad m_f(0) = \rho_f\,(\mathbf{V_F} - V_{u0}) \tag{11}$$

The integration ends when $m_f = 0$.

1.4 Assumptions

For this fluid system, we are using the same assumptions used in LRI's work:

Thermodynamics

- Nitrogen is an ideal gas with constant $\gamma = 1.4$ and $c_v = R/(\gamma - 1)$. Real-gas effects ($Z \approx 1.07$ at 3900 psig) are neglected.
- The pressurant tank expands isentropically: no heat transfer from the tank walls during the burn.
- The ullage is adiabatic: no heat transfer between the ullage gas and the tank walls or the fuel surface.
- Each gas volume is lumped, with a single uniform pressure and temperature.
- Joule–Thomson cooling is neglected.
- No mass transfer: no IPA vapor in the ullage, and no nitrogen dissolves into the fuel.

Flow and plumbing

- Each section of plumbing is described by a constant discharge coefficient times area, $C_d A$, independent of flow rate and Reynolds number (assuming turbulent flow).
- Flow is quasi-steady: line inertance and the volume of the lines are neglected, so flow rates respond instantly to pressure differences.
- The fuel is incompressible with constant density $\rho_f = 851$ kg/m^3.
- The pressurant line is treated as incompressible gas flow evaluated at the ullage density, valid because $\Delta P/P \approx 1\%$.
- The two $\frac{1}{2}''$ fuel branches are identical and discharge to the same pressure, so each carries half of the fuel flow.
- The engine is a constant-pressure boundary at $\mathbf{P_e} = 300$ psig.
- Hydrostatic head and vehicle acceleration are neglected.

Regulator

- The regulator works statically: no poppet inertia, spring dynamics, damping or dome volume. Its outlet pressure is a function of the current flow and inlet pressure only.
- Outlet pressure falls linearly with flow (Eq. (1)), which holds while the seat is choked ($P_{in} \gtrsim 1.9\, P_{out}$ absolute). This is true for the whole burn.
- The droop slope is interpolated from two manufacturer curves (Eq. (2)), held constant outside 1000–3000 psig inlet, and extrapolated linearly beyond the published 30 scfm range.
- The set pressure is constant: the supply pressure effect (outlet pressure shifting with inlet pressure) and the lockup pressure spike at zero flow are neglected.
- When fully open, the regulator is a fixed orifice with $C_d A = C_v/38$ in^2. This limit is not reached during the burn.

Operation

- The burn starts with the ullage at $\underline{P_{u0}}$ and both gas volumes at 300 K.
- The burn ends when the fuel mass reaches zero.

All constraints are relevant and taken from LRI's Propulsion Module designs. How this system is modeled is how the Propulsion Module is designed and what will be built (subject to change). How this model approximates this system is consistent with our previous tests and real data (especially the ullage pressure approximation).

2 The Method

This section analyzes and selects the numerical methods for our Initial Value Problem (IVP) following this structure.

- Section 2.1 restates the model as a standard IVP, explains the three operating regimes of the regulator that make up its right-hand side, and sets the accuracy we need.
- Section 2.2 derives the Jacobian of the IVP and its eigenvalues.
- Sections 2.3–2.5 use the Jacobian to analyze the truncation error, absolute stability and stiffness of the candidate methods.
- Section 2.6 compares the candidates on cost and selects one.
- Section 2.7 summarizes the resulting algorithm.

Each prediction is checked numerically. The eigenvalues, error measurements, step sizes and costs quoted here come from our method-studies code.

2.1 The initial value problem

The model in Section 1 can be written in the form,

$$\dot{u} = f(u), \qquad u(t_0) = u_0, \qquad u = \begin{bmatrix} m_p & T_p & m_u & T_u & m_f \end{bmatrix}^T, \tag{12}$$

with f given by Eqs. (7)–(10) and u_0 by Eq. (11). We march from $t_0 = 0$ to the burnout time t_f, defined by $m_f(t_f) = 0$. We will set the nominal configuration to $t_f = 27.918\,\text{s}$. As in Lecture 10, $t_k = t_0 + k\Delta t$, $u_k \approx u(t_k)$, and the global error is $e_k = u(t_k) - u_k$.

Note that the comparisons in the following sections use a fixed horizon $T_{\text{end}} = 27.6\,\text{s}$, just before burnout, so that every step size tested divides it evenly. The burnout step itself is treated in section 2.7.

Regulator operating regimes Each evaluation of f requires the nitrogen flow $\dot{m}_p$, which is set by the regulator. A spring-loaded regulator holds a poppet against a seat. This interaction causes the spring force corresponding to the set pressure P_{set}, and the outlet pressure P_{out}, which acts against it. Depending on how far the poppet is open, the regulator operates in one of the three regimes, and $\dot{m}_p$ has to be computed differently in each.

$$\dot{m}_p = \begin{cases} 0, & P_u \geq \min(P_{set}, P_p) \qquad \text{(closed)} \\ \dfrac{-K' + \sqrt{K'^2 + 4a\,(P_{set} - P_u)}}{2a}, & \text{otherwise, if valid} \qquad \text{(droop)} \\ \dot{m}_{\text{orif}}(P_{out}) = \dot{m}_{\text{line}}(P_{out}), & \text{otherwise} \qquad \text{(wide open)} \end{cases} \tag{13}$$

- **Closed (lock-up)** If the downstream pressure already meets or exceeds what the regulator can deliver, $P_u \geq \min(P_{set}, P_p)$, the poppet is seated and no gas flows. The nominal burn starts in this state, since $P_{u0} = P_{set}$.
- **Droop (regulating)** Once fuel leaves and P_u falls below P_{set}, the poppet opens just enough to pass the flow needed. A real regulator cannot hold $P_{out} = P_{set}$ while flowing. This means the more flow, the further the poppet must open and the lower the outlet pressure.

  This is the droop law, Eq. (1), $P_{out} = P_{set} - K'\dot{m}_p$, with slope $K' = K(P_p)/\rho_{std}$. The flow must also cross the pressurant line, which costs $P_{out} - P_u = \Delta p_\ell \equiv \dot{m}_p^2 R T_p/(2P_u C_d A_p^2)$ by Eq. (3). Eliminating P_{out} gives Eq. (9), whose positive root is the closed form (10) with $a = RT_p/(2P_u C_d A_p^2)$. The droop solution is valid only if the regulator could actually pass it, i.e. $P_{out} \leq P_p$ and the fully open valve at that P_{out} would carry at least $\dot{m}_p$.

- **Wide-open (saturated)** If the droop solution is not valid, typically when the tank pressure P_p has fallen too low, the poppet is fully open and the regulator acts as a fixed orifice with $C_d A_{reg} = C_v/38\,\mathrm{in}^2$. The flow through it is the compressible orifice flow, which is choked for $r \le r_c$.

$$\dot{m}_{\mathrm{orif}}(P_{out}) = C_d A_{reg}\, P_p \sqrt{\frac{2\gamma}{(\gamma-1)RT_p}\Big(r^{2/\gamma} - r^{(\gamma+1)/\gamma}\Big)}, \quad r = \max\Big(\frac{P_{out}}{P_p},\, r_c\Big), \quad r_c = \Big(\frac{2}{\gamma+1}\Big)^{\frac{\gamma}{\gamma-1}}, \tag{14}$$

$\dot{m}_{\mathrm{line}}(P_{out}) = C_d A_p \sqrt{2P_u(P_{out} - P_u)/(RT_p)}$

It must equal the line flow from Eq. (3). That single equation is solved for $P_{out} \in (P_u, P_p)$ by a bracketing root-finder.

According to the method-studies code, the nominal burn is in the closed regime only at $t = 0$. It enters the droop regime immediately and stays there, and the wide-open limit is never reached. The regimes still matter for the method, for two reasons. First, f is continuous across the switches but its derivatives are not (discussed in section 2.2 and 2.3). Second, a parameter sweep can push a configuration into a different regime.

Accuracy target The quantity the engine sees, and the one our design questions are about, is the fuel flow $\dot{m}_f$. For this project, we require the method to deliver it to within 1% over the whole burn. This can be expressed as follows.

$$e_{\dot{m}_f} \equiv \max_k \frac{\big|\dot{m}_f(u_k) - \dot{m}_f(u(t_k))\big|}{\dot{m}_f(u(t_k))} \;\le\; 10^{-2} \tag{15}$$

Cost The model is meant to be run many times as a design tool, so we measure cost as the number of evaluations of f per run, plus evaluations of the Jacobian for implicit methods, together with the wall-clock time. Each evaluation of f includes the regime logic of (13).

2.2 Jacobian and eigenvalues

Following Lecture 12, a small perturbation $\tilde{u}$ about a reference state $\bar{u}$ evolves as $\dot{\tilde{u}} \approx J\tilde{u}$, where $J = \partial f/\partial u$ is evaluated at $\bar{u}$. Our system has no rest state during the burn, since fuel is always flowing. We therefore take $\bar{u} = u(t^*)$ at a sequence of states the solution passes through, and use the one with the largest $|\lambda|$ to set the stability limit.

Analytic Jacobian Every term of f is an explicit function of the state through five intermediate quantities, P_p, V_u, P_u, $\dot{m}_f$ and $\dot{m}_p$ [Eqs. (8a), (8b), (13)], so J follows from the chain rule. We write gradients as row vectors, $\partial z/\partial u = [\partial_{m_p}, \partial_{T_p}, \partial_{m_u}, \partial_{T_u}, \partial_{m_f}](z)$, and e_j^T for the j-th unit row.

For Pressures and fuel flow. From the ideal-gas law and $V_u = V_F - m_f/\rho_f$ (so $\partial V_u/\partial m_f = -1/\rho_f$).

$$\frac{\partial P_p}{\partial u} = \frac{P_p}{m_p}\, e_1^T + \frac{P_p}{T_p}\, e_2^T, \tag{16}$$

$$\frac{\partial P_u}{\partial u} = \frac{P_u}{m_u}\, e_3^T + \frac{P_u}{T_u}\, e_4^T + \frac{P_u}{\rho_f V_u}\, e_5^T, \tag{17}$$

$$\frac{\partial \dot{m}_f}{\partial u} = \frac{\dot{m}_f}{2(P_u - P_e)}\, \frac{\partial P_u}{\partial u} \tag{18}$$

For Nitrogen flow (droop regime). Rather than differentiate the closed form (10), it is simpler to differentiate the equation it solves. Write Eq. (9) as follows.

$$F(\dot{m}_p;\, P_u, T_p, P_p) \equiv P_{set} - K'(P_p)\,\dot{m}_p - \frac{\dot{m}_p^2 RT_p}{2P_u C_d A_p^2} - P_u = 0 \tag{19}$$

Along any change of state F stays zero, so $dF = F_{\dot m}\,d\dot m_p + F_{P_u}dP_u + F_{T_p}dT_p + F_{P_p}dP_p = 0$, with

$$F_{\dot m} = -\Big(K' + \frac{2\Delta p_\ell}{\dot m_p}\Big) \equiv -D, \qquad F_{P_u} = \frac{\Delta p_\ell}{P_u} - 1, \qquad F_{T_p} = -\frac{\Delta p_\ell}{T_p}, \qquad F_{P_p} = -\dot m_p\,K'_P, \tag{20}$$

where $K'_P = dK'/dP_p$ is the slope of the K interpolation, Eq. (2): $(K_2 - K_1)/[\rho_{std}(P_2 - P_1)]$ between the two curves and zero outside them. Solving for $d\dot m_p$ yields the following.

$$\frac{\partial \dot m_p}{\partial u} = -\frac{1}{D}\left[\Big(1 - \frac{\Delta p_\ell}{P_u}\Big)\frac{\partial P_u}{\partial u} + \frac{\Delta p_\ell}{T_p}\,e_2^T + \dot m_p\,K'_P\,\frac{\partial P_p}{\partial u}\right]. \tag{21}$$

The three terms say that the regulator flow drops if the ullage pressure rises, if the pressurant warms, or if the tank pressure drops in the range where droop K' grows. In the closed regime $\dot m_p \equiv 0$, so $\partial\dot m_p/\partial u = 0$. The wide-open branch is not reached on the nominal burn so it is not needed here.

Finally, for the rows of J. We can now differentiate Eqs. (7a)–(7e) with (16)–(21), where $f_4 = \big[\dot m_p(\gamma T_p - T_u) - P_u\dot m_f/(\rho_f c_v)\big]/m_u$.

$$\frac{\partial f_1}{\partial u} = -\frac{\partial \dot m_p}{\partial u} \tag{22}$$

$$\frac{\partial f_2}{\partial u} = -(\gamma - 1)\left[\frac{T_p}{m_p}\frac{\partial \dot m_p}{\partial u} + \frac{\dot m_p}{m_p}\,e_2^T - \frac{T_p\dot m_p}{m_p^2}\,e_1^T\right] \tag{23}$$

$$\frac{\partial f_3}{\partial u} = +\frac{\partial \dot m_p}{\partial u} \tag{24}$$

$$\frac{\partial f_4}{\partial u} = \frac{1}{m_u}\left[(\gamma T_p - T_u)\frac{\partial \dot m_p}{\partial u} + \dot m_p\big(\gamma\,e_2^T - e_4^T\big) - \frac{1}{\rho_f c_v}\Big(\dot m_f + P_u\frac{\partial \dot m_f}{\partial P_u}\Big)\frac{\partial P_u}{\partial u} - f_4\,e_3^T\right] \tag{25}$$

$$\frac{\partial f_5}{\partial u} = -\frac{\partial \dot m_f}{\partial u} \tag{26}$$

This gives J as a 5×5 matrix whose rows are (22)–(26). Two structural facts follow directly. Rows 1 and 3 are negatives of each other, which is nitrogen mass conservation. On the nominal burn $P_p > 3000$ psig throughout, so $K'_P = 0$, and m_p enters J only through the last term of (23). As for an extra check, the method-studies code compares (22)–(26) with a central-difference Jacobian of f at 25 states along the burn; the two agree to within 3×10^{-10}.

Eigenvalues Table 2 and Fig. 3 give the five eigenvalues of J along the nominal solution. All five are real.

Table 2: Eigenvalues of J (s^{-1}) along the nominal solution. λ_4 is zero to round-off ($|\lambda_4| < 10^{-9}$).

t^* (s)	λ_1	λ_2	λ_3	λ_4	λ_5
0^+	-6.52	-0.099	-6.7×10^{-6}	0	≈ 0
0.475	-8.53	-0.165	-2.05×10^{-3}	0	9.2×10^{-5}
1	-7.76	-0.148	-2.09×10^{-3}	0	9.7×10^{-5}
5	-4.12	-0.080	-2.13×10^{-3}	0	9.8×10^{-5}
10	-2.60	-0.051	-2.18×10^{-3}	0	1.01×10^{-4}
20	-1.51	-0.030	-2.29×10^{-3}	0	1.08×10^{-4}
27.6	-1.16	-0.022	-2.38×10^{-3}	0	1.14×10^{-4}

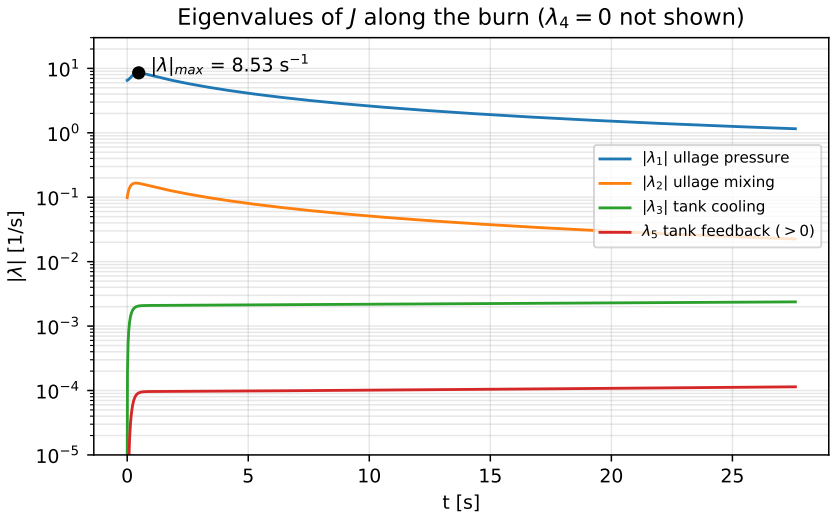


Figure 3: Magnitude of the eigenvalues of J along the nominal burn. The fastest mode peaks at $|\lambda|_{\max} = 8.53\,\mathrm{s}^{-1}$ at $t^* = 0.475\,\mathrm{s}$.

Each eigenvalue corresponds to a physical mode of the feed system.

- λ_1**, ullage pressure** A perturbation of P_u is corrected by the regulator, which admits more gas when P_u drops, and by the fuel flow, which drains the ullage faster when P_u rises. Both act through the ullage volume, so the mode is fastest early in the burn, when V_u is small ($|\lambda_1|_{\max} = 8.53\,\mathrm{s}^{-1}$ at $t^* = 0.475\,\mathrm{s}$), and slows about sevenfold as V_u grows from 1.8 to 17.1 L. Its time scale is $1/|\lambda_1| = 0.12$–$0.87\,\mathrm{s}$, and it is the start-up transient. In other words, P_u falls from 360 to 309 psig, and $\dot{m}_f$ from 1.23 to 0.47 kg/s, within about 0.5 s.

- λ_2**, ullage mixing** Nitrogen enters the ullage at stagnation temperature γT_p, so a perturbation of T_u decays as fresh gas replaces the ullage contents; time scale 6–45 s.

- λ_3 **and** λ_5**, pressurant tank** $\lambda_3 \approx -(\gamma - 1)\dot{m}_p/m_p$ is the isentropic cooling of the tank as it empties (time scale 420–490 s). $\lambda_5 \sim 10^{-4}\,\mathrm{s}^{-1}$ is positive: a tank holding slightly more gas cools more slowly, its warmer gas passes the line slightly less easily, and so it keeps even more gas. Over the whole burn $\lambda_5 t_f < 3.2 \times 10^{-3}$, so this physical growth is below 0.4%.

- $\lambda_4 = 0$**, nitrogen mass** Since $\dot{m}_p + \dot{m}_u = 0$ [Eqs. (7a), (7c)], $c = [1, 0, 1, 0, 0]^T$ satisfies $c^T J = 0$. A perturbation that adds nitrogen to the system neither grows nor decays.

2.3 The method and its truncation error

Derivation of RK4 As in Lecture 10, integrating the IVP (6) across one step gives the following exact relation.

$$u(t_{k+1}) - u(t_k) = \int_{t_k}^{t_{k+1}} f\big(u(t)\big)\,dt \tag{27}$$

Every method approximates the integral on the right. The RK4 method is implemented as follows.

$$\int_{t_k}^{t_{k+1}} f\,dt \approx \frac{\Delta t}{6}\Big[f_k + 4f_{k+1/2} + f_{k+1}\Big]. \tag{28}$$

The states at the midpoint and at t_{k+1} are unknown, so RK4 estimates the slope there with explicit predictions.

$$
\begin{aligned}
y_1 &= f(u_k) && \text{slope at } t_k,\\
y_2 &= f\big(u_k + \tfrac{1}{2}\Delta t\, y_1\big) && \text{slope at the midpoint, predicted with } y_1,\\
y_3 &= f\big(u_k + \tfrac{1}{2}\Delta t\, y_2\big) && \text{slope at the midpoint, re-predicted with } y_2,\\
y_4 &= f\big(u_k + \Delta t\, y_3\big) && \text{slope at } t_{k+1},\ \text{predicted with } y_3,
\end{aligned}
\tag{29}
$$

$$
u_{k+1} = u_k + \frac{\Delta t}{6}\big(y_1 + 2y_2 + 2y_3 + y_4\big). \tag{30}
$$

The midpoint weight 4/6 of (28) is split equally between the two midpoint estimates y_2 and y_3. Every argument on the right is known when it is used, so the method is explicit and costs four evaluations of f per step.

Truncation error Following Lecture 11, the truncation error is computed as follows.

$$
\tau_k = \frac{u(t_{k+1}) - u(t_k)}{\Delta t} - \frac{1}{6}\Big(y_1 + 2y_2 + 2y_3 + y_4\Big)\Big|_{u_k = u(t_k)} \tag{31}
$$

For the Adams methods example in Lecture 11, we expanded τ_k once for the whole family. Unfortunately, RK4 cannot be treated that way, because its stages evaluate f at predicted states rather than at grid values. For a general nonlinear f the expansion produces many cross terms between the stages; a complete treatment shows that RK4's coefficients satisfy all eight conditions required for fourth order. Rather than reproduce that algebra, we examine two special cases. Each isolates one of the two ingredients of the method, and each can be expanded by hand.

1. *The quadrature (where f is sampled and how the samples are weighted).* If $f = f(t)$ does not depend on u, the predictions inside the stages have no effect: $y_1 = f(t_k)$, $y_2 = y_3 = f(t_k + \Delta t/2)$, $y_4 = f(t_{k+1})$, and (30) reduces to exactly Simpson's rule (28). Its error follows from Taylor-expanding about the midpoint $m = t_k + h$, with $h = \Delta t/2$. Odd powers of h cancel by symmetry.

$$
\int_{m-h}^{m+h} f\, dt = 2hf(m) + \frac{h^3}{3}f''(m) + \frac{h^5}{60}f^{(4)}(m) + \cdots \tag{32}
$$

$$
\frac{h}{3}\big[f(m-h) + 4f(m) + f(m+h)\big] = 2hf(m) + \frac{h^3}{3}f''(m) + \frac{h^5}{36}f^{(4)}(m) + \cdots \tag{33}
$$

The first two terms agree, so Simpson's rule is exact for cubics. The difference is $\big(\frac{1}{60} - \frac{1}{36}\big)h^5 f^{(4)} = -\frac{h^5}{90}f^{(4)} = -\frac{\Delta t^5}{2880}f^{(4)}$. Dividing by Δt, and using $f^{(4)} = u^{(5)}$,

$$
\tau_k = -\frac{\Delta t^4}{2880}\, u^{(5)}(m) + O(\Delta t^6). \tag{34}
$$

2. *The predictions (how the stages feed into one another).* If $f = \lambda u$, the quadrature plays no special role, but each stage now depends on the one before. Substituting into (29) with $w = \Delta t\lambda$, each stage adds one more power of w:

$$
y_1 = \lambda u_k, \tag{35}
$$

$$
y_2 = \lambda\big(u_k + \tfrac{1}{2}\Delta t\, y_1\big) = \lambda\big(1 + \tfrac{1}{2}w\big)u_k, \tag{36}
$$

$$
y_3 = \lambda\big(u_k + \tfrac{1}{2}\Delta t\, y_2\big) = \lambda\big(1 + \tfrac{1}{2}w + \tfrac{1}{4}w^2\big)u_k, \tag{37}
$$

$$
y_4 = \lambda\big(u_k + \Delta t\, y_3\big) = \lambda\big(1 + w + \tfrac{1}{2}w^2 + \tfrac{1}{4}w^3\big)u_k. \tag{38}
$$

Inserting these into (30),

$$u_{k+1} = u_k + \frac{w}{6}\Big(6 + 3w + w^2 + \tfrac{1}{4}w^3\Big)u_k = \underbrace{\Big(1 + w + \tfrac{1}{2}w^2 + \tfrac{1}{6}w^3 + \tfrac{1}{24}w^4\Big)}_{R(w)}\, u_k. \tag{39}$$

The exact solution multiplies by $e^w = 1 + w + \frac{1}{2}w^2 + \frac{1}{6}w^3 + \frac{1}{24}w^4 + \frac{1}{120}w^5 + \cdots$ over the same step. $R(w)$ matches this term by term through w^4 and first differs at w^5, so $e^w - R(w) = \frac{1}{120}w^5 + O(w^6)$. Substituting the exact solution into (31),

$$\tau_k = \frac{e^w - R(w)}{\Delta t}\, u(t_k) = \frac{\Delta t^4}{120}\, \lambda^5 u(t_k) + O(\Delta t^5) = \frac{\Delta t^4}{120}\, u^{(5)}(t_k) + O(\Delta t^5). \tag{40}$$

Both cases give $\tau_k = O(\Delta t^4)$, multiplying a fifth derivative of the solution. Only the constant differs (1/2880 versus 1/120), because the constant depends on the problem while the power of Δt is a property of the method. Case 1 shows that the weights $\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6}$ are right. Case 2 shows that the chain of predictions is also right. A general nonlinear f combines both effects, and the order conditions guarantee that no lower power of Δt survives. Since RK4 is stable as $\Delta t \to 0$ (discussed in section 2.4), the global error converges at the same rate, $e_k = O(\Delta t^4)$.

Other candidates The orders of the other methods considered follow from Lecture 10, 11 and 13: forward and backward Euler $p = 1$; Heun, trapezoid, AB2 and BDF2 $p = 2$; AB4 $p = 4$.

Numerical confirmation Figure 4(a) shows the truncation error measured on our IVP. For each method it takes one step from the exact solution at $t_k = 2\,\text{s}$, which we approximate by a reference RK4 solution with $\Delta t_{ref} = 0.4/2^{10}\,\text{s}$. Figure 4(b) shows the global error $e_{\dot{m}_f}$ over the whole burn. The fitted slopes (Table 3) match the predicted orders to within 0.04 for every method. The estimated error of the reference solution in $\dot{m}_f$ is 3×10^{-13}, well below every error measured.

Table 3: Predicted order p and measured slopes of τ_k and $e_{\dot{m}_f}$ (least-squares fit over the three smallest Δt).

	FE	Heun	RK4	AB2	AB4	BE	Trap	BDF2
p (theory)	1	2	4	2	4	1	2	2
slope of τ_k	1.00	1.99	3.99	2.00	3.97	0.96	1.97	1.97
slope of $e_{\dot{m}_f}$	1.01	2.02	4.02	2.00	3.98	0.98	2.00	1.99

Panel (a) is smooth for every method at every step size, including steps at which the method is unstable. This means τ_k is a one-step quantity and knows nothing about stability.

Panel (b) follows the predicted slope only while the step is below the method's stability limit. Past it, the error leaves the line, as AB4 does sharply beyond $\Delta t \approx 0.04$ s.

Smoothness requirement Every expansion above assumes $u(t)$ has enough continuous derivatives on each step ($u^{(5)}$ for a fourth-order method). Our f is only piecewise smooth, meaning its derivatives jump at the regulator switches of (13) and at the 1000 and 3000 psig breakpoints of $K(P_p)$.

On the nominal burn the solution leaves the closed regime at $t = 0$ and then stays in the droop regime with $P_p \in [3141, 3900]$ psig, so f is smooth along the solution for $t > 0$. The measured slope of 4.02 for RK4 confirms that the formal order holds. A configuration whose trajectory crosses a switch during the burn would place the kink inside one step. That step's local error is then no longer $O(\Delta t^{p+1})$, and the measured rate could fall below p.

2.4 Absolute stability

Following Lecture 12, we apply each method to the model problem $\dot{u} = \lambda u$, with λ an eigenvalue of J and $w = \Delta t\lambda$. A one-step method gives $u_{k+1} = R(w)u_k$ and is absolutely stable where $|R(w)| < 1$. A multi-step method is absolutely stable where every root ζ of its characteristic polynomial has $|\zeta| < 1$. A march is stable only if every $w_j = \Delta t\lambda_j$ lies in the method's region.

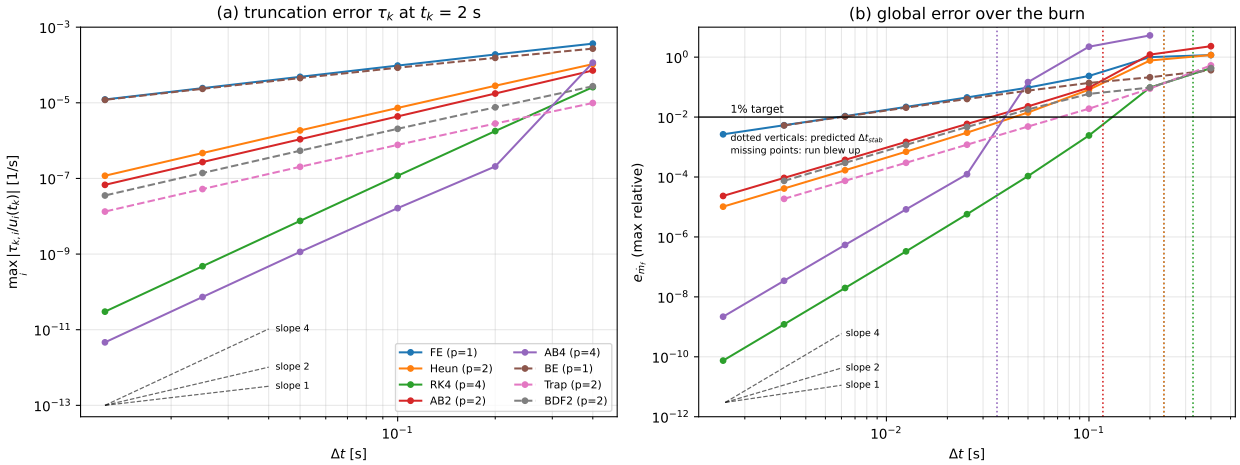


Figure 4: (a) Truncation error τ_k at $t_k = 2\,\mathrm{s}$ and (b) global error $e_{\dot{m}_f}$ over the burn, versus Δt. Dashed fans show slopes 1, 2 and 4. In (b), dotted verticals mark each explicit method's predicted stability limit Δt_{stab} (section 2.4), and the horizontal line is the 1% target.

RK4 The amplification factor was already derived in lecture 12.

$$R(w) = 1 + w + \tfrac{1}{2}w^2 + \tfrac{1}{6}w^3 + \tfrac{1}{24}w^4. \tag{41}$$

Because every eigenvalue in Table 2 is real, only the real axis matters. $R(w)$ is the degree-4 Taylor polynomial of e^w, which is positive for every real w, so $R(w) = -1$ has no real solution. The region therefore ends on the negative real axis where $R(w) = 1$.

$$w\left(1 + \tfrac{1}{2}w + \tfrac{1}{6}w^2 + \tfrac{1}{24}w^3\right) = 0 \quad \Longrightarrow \quad w^* = 2.785, \tag{42}$$

RK4 is therefore stable for $-2.785 < w < 0$. Figure 5(a) shows the full region in the complex plane.

Other candidates The same procedure in Lecture 12 gives $w^* = 2$ for forward Euler and Heun, $w^* = 1$ for AB2 and $w^* = 3/10$ for AB4. Backward Euler, trapezoid and BDF2 contain the whole negative real axis (no w^*). The method-studies code recomputes w^* for every method directly from $R(w)$ or the characteristic roots, and reproduces these values to four digits.

Stability limit for our IVP The decaying modes require $\Delta t|\lambda_j| < w^*$, and the fastest mode is the binding one:

$$\Delta t < \Delta t_{\mathrm{stab}} = \frac{w^*}{|\lambda|_{\max}}, \qquad |\lambda|_{\max} = 8.53\ \mathrm{s}^{-1}. \tag{43}$$

For RK4, $\Delta t_{\mathrm{stab}} = 2.785/8.53 = 0.326\,\mathrm{s}$; the other methods are listed in Table 4. Because $|\lambda_1|$ decays to $1.16\,\mathrm{s}^{-1}$ by burnout, the limit is tightest early in the burn and relaxes to 2.4 s at the end. In Fig. 5 the RK4 operating points $w_j = \Delta t\lambda_j(t)$, taken at the step that meets the 1% target ($\Delta t = 0.133\,\mathrm{s}$), all lie inside its region.

The two non-decaying modes need no restriction. For $\lambda_4 = 0$, $R(0) = 1$: the method keeps the perturbation, as the true solution does. For $\lambda_5 > 0$, $|w_5| < 4 \times 10^{-5}$ at any step we use, so $R(w_5) \approx e^{w_5}$ reproduces the small physical growth.

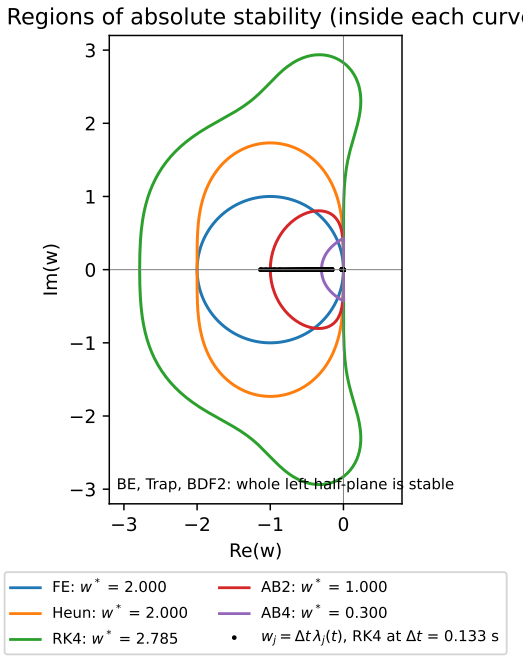


Figure 5: Regions of absolute stability of the explicit candidates, with RK4's operating points $w_j = \Delta t\lambda_j(t)$ at $\Delta t = 0.133\,\mathrm{s}$ for every state along the burn (black, on the negative real axis).

AB4 Table 4 shows AB4 meeting the 1% target at $\Delta t = 0.0453\,\mathrm{s}$, about 30% beyond its predicted $\Delta t_{\mathrm{stab}} = 0.035\,\mathrm{s}$. This does not contradict the theory. At that step the fastest mode lies outside AB4's region only while $|\lambda_1|$ is near its peak ($0.04 < t < 1.76\,\mathrm{s}$), so the growth acts for less than two seconds. The instability is still visible in Fig. 4(b), which can be seen in AB4's error follows Δt^4 up to $\Delta t \approx 0.035\,\mathrm{s}$, and a 12% larger step, from 0.0406 to 0.0453 s, then raises it elevenfold, from 9.2×10^{-4} to 9.9×10^{-3}. AB4's 1% point therefore lies on the instability wall, not on its Δt^4 error curve.

2.5 Stiffness

Lecture 13 calls an IVP stiff when it contains a mode that is unimportant to the dynamics but imposes the stability restriction on the step. It is measured by the stiffness ratio. Taking the decaying modes,

$$R_s = \frac{\max_j |\lambda_j|}{\min_j |\lambda_j|} = \frac{|\lambda_1|}{|\lambda_3|} \approx \frac{8.53}{2.05 \times 10^{-3}} \approx 4.2 \times 10^3 \quad (t^* = 0.475\ \mathrm{s}), \tag{44}$$

falling to about 490 by the end of the burn. By this measure the IVP is stiff. Lecture 13 reads R_s as roughly how many more steps stability demands than accuracy. It says the slowest mode sets how long we must march, and the fastest sets how small the step must be, but neither part holds here.

- The fast mode is not unimportant. λ_1 is the start-up transient of the ullage pressure, which sets the fuel flow the engine receives; resolving it is part of the answer.
- The slowest mode does not set how long we march. Its time scale, $1/|\lambda_3| \approx 490\,\mathrm{s}$, is much longer than the burn, which ends when the fuel runs out at $t_f = 27.9\,\mathrm{s}$.

We therefore count the steps directly. Over the study horizon, stability alone requires

$$N_{\mathrm{stab}} = \frac{T_{\mathrm{end}}}{\Delta t_{\mathrm{stab}}} = \frac{T_{\mathrm{end}}\,|\lambda|_{\max}}{w^*} \approx \frac{236}{w^*} \tag{45}$$

steps. Accuracy alone requires $N_{\text{acc}} = T_{\text{end}}/\Delta t_{\text{acc}}$, where Δt_{acc} is the step at which the method's error line in Fig. 4(b), $e_{\dot{m}_f} \approx C\Delta t^p$ fitted at small Δt, reaches 1%. This ignores stability on purpose.

	FE	Heun	RK4	AB2	AB4
N_{stab}	118	118	85	236	785
N_{acc}	4713	568	165	853	381
step set by	accuracy	accuracy	accuracy	accuracy	stability

For every explicit candidate except AB4, $N_{\text{stab}} < N_{\text{acc}}$: the stability wall lies at a larger step than accuracy already asks for, so it costs nothing. For RK4, stability alone would allow about half the steps accuracy needs. (The measured 1% step of Table 4, 0.133 s, is somewhat smaller than the fitted 0.167 s, because near 1% the RK4 error is just leaving its asymptotic slope. It is still 2.5 times below Δt_{stab}.) AB4 is the exception, because its region is so small ($w^* = 0.3$). So although the IVP is stiff by R_s, it is not stiff in the sense that matters for choosing a method.

2.6 Cost and choice of method

Cost per step An explicit one-step method with s stages costs s evaluations of f per step (1 for forward Euler, 2 for Heun, 4 for RK4). An Adams–Bashforth method costs one new evaluation per step, plus a one-step starter (we use RK4) for its first $r-1$ steps. An implicit method must solve a nonlinear system for u_{k+1} at every step. For backward Euler, for example, this is $G(z) = z - u_k - \Delta t\, f(z) = 0$. Newton's method solves it by iterating $(I - \Delta t\, J)\,\delta z = -G(z)$, which costs one evaluation of f, one of J and one 5×5 linear solve per iteration. We solve these systems with `scipy.optimize.fsolve`, supplying the analytic Jacobian of section 2.2.

Comparison Table 4 collects order, stability limit and measured cost for every candidate. Each row comes from an actual run at the largest step that meets the 1% target, found by bisection over step sizes that are integer multiples of Δt_{ref}, i.e. to within 3.9×10^{-4} s. Figure 6 shows the error reached against the work spent, for every step size tested.

Table 4: Candidate methods on the nominal case, each at the largest step meeting $e_{\dot{m}_f} \leq 1\%$, over $T_{\text{end}} = 27.6$ s. Implicit methods also evaluate the Jacobian (J-evals). Wall times are from a single desktop run and are indicative only.

Method	p	w^*	Δt_{stab} (s)	$\Delta t_{1\%}$ (s)	steps	f-evals	J-evals	wall time (s)
Forward Euler	1	2	0.234	0.0055	5046	5046	–	0.17
Heun	2	2	0.234	0.0426	648	1296	–	0.033
RK4	4	2.785	0.326	**0.133**	**207**	828	–	**0.017**
AB2	2	1	0.117	0.0324	851	856	–	0.036
AB4	4	0.3	0.035	0.0453*	609	622	–	0.029
Backward Euler	1	none	none	0.0059	4710	72189	13982	2.4
Trapezoid	2	none	none	0.0723	381	2448	780	0.11
BDF2	2	none	none	0.0367	751	10409	2069	0.44

*Beyond Δt_{stab}, on the steep instability branch of AB4's error curve (Sec. 2.4).

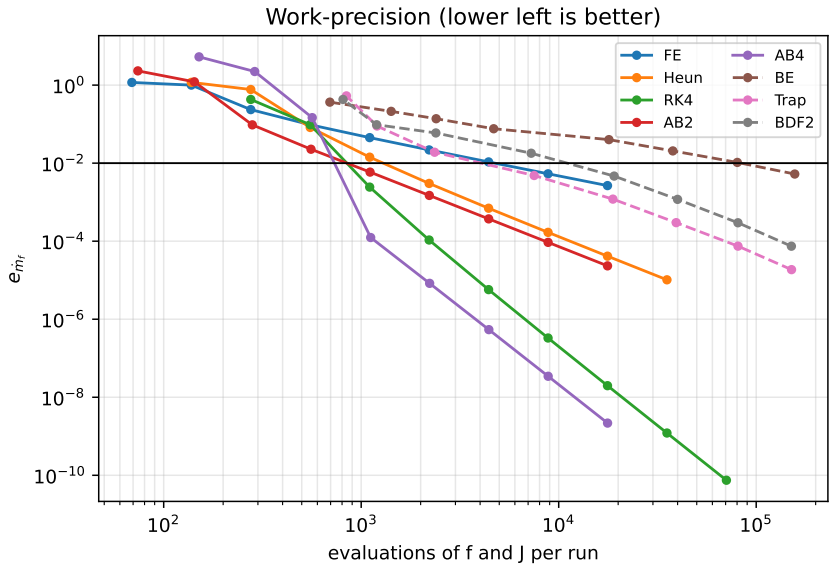


Figure 6: Work-precision: error $e_{\dot{m}_f}$ against evaluations of f and J per run, for every step size tested. The horizontal line is the 1% target.

How the measurements compare with our expectations Most of the predictions of sections 2.3–2.5 are confirmed directly. Every method converges at its predicted order, every w^* is reproduced, and every explicit method except AB4 is accuracy-limited. Three results differ from what we expected, and we state them explicitly because none of them changes the choice.

- *AB4 runs past its stability limit.* Its 1% step is 0.0453 s, which is not $\leq$ 0.035s. As explained in section 2.4, this is because $|\lambda_1|$ is large only briefly. It still sits on the instability wall, so the conclusion that AB4 is stability-limited holds.
- *Implicit methods are cheaper per step than a finite-difference Newton solve would make them*, which would spend five extra evaluations of f on every Jacobian. With the analytic Jacobian of section 2.2, the trapezoid method needs about 6.4 evaluations of f and 2.0 of J per step, and BDF2 and backward Euler about 14–15 of f and 3 of J. This is still more than RK4's four evaluations, for no gain in step size, and in wall time the implicit methods are 3–14 times slower than the explicit method of the same order and 6–140 times slower than RK4.
- *RK4's 1% step is smaller than the asymptotic estimate.* The fit $e \approx C\Delta t^4$ predicts 0.167 s, but the measured 1% step is 0.133 s. Near 0.13 s the error is just leaving its asymptotic range and grows faster than Δt^4. The measured step is still 2.5 times below the stability limit.

Choice Table 4 and Fig. 6 support the following conclusions.

- *Implicit methods buy nothing here.* Because accuracy sets the step (section 2.5), they need about as many steps as explicit methods of the same order: trapezoid 0.072 s against Heun 0.043 s; BDF2 0.037 s against AB2 0.032 s. Each step, however, carries a nonlinear solve.
- *First-order methods are too expensive.* Forward and backward Euler need about 5000 steps to reach 1% on $\dot{m}_f$, and their error falls only in proportion to Δt.
- *Among the explicit methods of order ≥ 2, RK4 and AB4 are the strongest.* When it is stable, AB4 reaches a given error with fewer evaluations of f than RK4 (Fig. 6), because it reuses past values instead of computing new stages; at the 1% target it needs 622 evaluations against RK4's 828. Its

weakness is entirely stability. Its usable step is set by $|\lambda|_{\max}$, and its 1% point sits on the instability wall. $|\lambda|_{\max}$ changes with the design inputs: by section 2.2 it grows with a smaller initial ullage volume or a regulator with a smaller droop slope K. So a step that works for the nominal case can fail for another configuration in the parameter sweep, abruptly rather than gradually. AB4 also needs a starter, and its stored history has to be restarted at the burnout step.

- *RK4* needs the fewest steps of any method (207), has the lowest wall time in our runs (0.017 s), and runs with a factor of 2.5 margin below its stability limit. A configuration with a faster ullage mode therefore stays stable at the same step. It is self-starting, so the shortened last step at burnout (section 2.7) needs no special treatment. Its error falls as Δt^4: halving the step cuts the error by 16, which makes the tight-tolerance reference solutions of section 3 cheap.

We therefore select the fourth-order Runge–Kutta method, Eqs. (29)–(30), with a fixed step. Any $\Delta t \leq 0.133\,\mathrm{s}$ meets the 1% target on the nominal case; the step used for the results is chosen in Sec. 3.

2.7 Algorithm

The method advances the solution from t_k to $t_{k+1} = t_k + \Delta t$ as follows. The evaluation of f (step 2) is the same for every stage.

1. **Start** Set u_0 from Eq. (11), $t_0 = 0$, and choose Δt (Sec. 3).
2. **Evaluate** $f(u)$ for a given state $u = [m_p, T_p, m_u, T_u, m_f]$:
   (a) pressures and volume from Eq. (8a): $P_p = m_p R T_p / V_p$, $V_u = V_F - m_f/\rho_f$, $P_u = m_u R T_u / V_u$;
   (b) fuel flow from Eq. (8b): $\dot{m}_f = C_d A_F \sqrt{2\rho_f (P_u - P_e)}$;
   (c) regulator flow $\dot{m}_p$ from (13): if $P_u \geq \min(P_{set}, P_p)$, set $\dot{m}_p = 0$ (closed); otherwise compute the droop root (10) and keep it if it is valid; otherwise solve $\dot{m}_{\mathrm{orif}} = \dot{m}_{\mathrm{line}}$ for P_{out} (wide open);
   (d) return the derivatives of Eqs. (7a)–(7e).
3. **RK4 step** Compute the four stages (29), each with one evaluation of step 2, and form the trial state $u_{k+1} = u_k + \frac{\Delta t}{6}(y_1 + 2y_2 + 2y_3 + y_4)$.
4. **Check for burnout** If $m_{f,k+1} > 0$, accept the step, set $t_{k+1} = t_k + \Delta t$ and return to step 3.
5. **Last step** Otherwise the fuel runs out inside the step. Find the step length $h \in (0, \Delta t]$ at which $m_f = 0$, by Newton's method on $g(h) = m_f$ after an RK4 step of length h from u_k. Since $dm_f/dt = -\dot{m}_f$, $g'(h) \approx -\dot{m}_f$:

$$h^{(0)} = \frac{m_{f,k}}{\dot{m}_f(u_k)}, \qquad h^{(i+1)} = h^{(i)} + \frac{g(h^{(i)})}{\dot{m}_f\big(u(h^{(i)})\big)}. \tag{46}$$

   The first guess assumes constant fuel flow over the last step. Stop when $|m_f| < 10^{-10} m_{f,0}$, set $t_f = t_k + h$, and end.

Each regular step costs four evaluations of f, and the last step costs four per Newton iteration. Because m_f is nearly linear in time over one step, the method-studies code needs at most two iterations at every step size tested. Applied to the nominal burn, the algorithm gives $t_f = 27.917985$ s at Δt_{ref}. The burnout time converges at fourth order: its error is 1.3×10^{-4} s at $\Delta t = 0.1$ s, 5.8×10^{-6} s at 0.05 s and 3.1×10^{-7} s at 0.025 s. In every case $m_f(t_f)$ is below 2×10^{-11} kg.

Summary of the method The feed-system IVP has real eigenvalues, with a fast ullage-pressure mode ($|\lambda|_{\max} = 8.53\,\mathrm{s}^{-1}$) that is also the start-up transient we need to resolve. The IVP is stiff by R_s, but the 1% accuracy target on $\dot{m}_f$ needs a step that is already inside the stability limit of RK4, so an implicit method would add a nonlinear solve per step without allowing a larger step. RK4 is fourth-order accurate, has the largest stability interval of the explicit candidates, needs no starter, and meets the target in the fewest steps and least time. It is applied with a fixed step and a root-found final step at burnout.

3 Correct Implementation

To ensure that our custom, from-scratch integration loop produces mathematically and physically valid results, we conducted a series of error studies and a parameter evaluation. These checks guarantee that our chosen numerical method—the fourth-order Runge-Kutta (RK4) method—converges at the expected theoretical rate, perfectly adheres to the underlying thermodynamic equations, and operates within the bounds of stability and required accuracy.

3.1 Error Studies

Our error studies evaluate both the mathematical convergence of the integration method and the physical correctness of the formulated equations of motion.

Convergence Rate Verification: We evaluated the global error over the complete engine burn by comparing integrations at various time steps against a high-resolution reference solution computed at $\Delta t_{ref} = 0.4/2^{10}$ s. The global error was defined as the maximum relative error in the fuel mass flow rate, $e_{\dot{m}_f}$, over the horizon. As shown in Figure 4, the error for our custom RK4 method decreases precisely parallel to the theoretical $\mathcal{O}(\Delta t^4)$ slope. A least-squares fit of the error data for the smallest step sizes yields an observed convergence rate of $p = 4.02$, confirming that the explicit integrator is implemented correctly and achieves its formal theoretical accuracy.

Verification of Correct Equations: An integrator can cleanly converge to the wrong physics if the right-hand side equations, $f(u)$, are malformed. To prove our equations are structurally correct, we tested our numerical solution against a fundamental conservation law: the total mass of the nitrogen pressurant gas. Because nitrogen solely flows from the pressurant tank to the fuel tank ullage without leaving the system, the sum of the masses, $m_{N2} = m_p(t) + m_u(t)$, must remain strictly equal to its initial value.

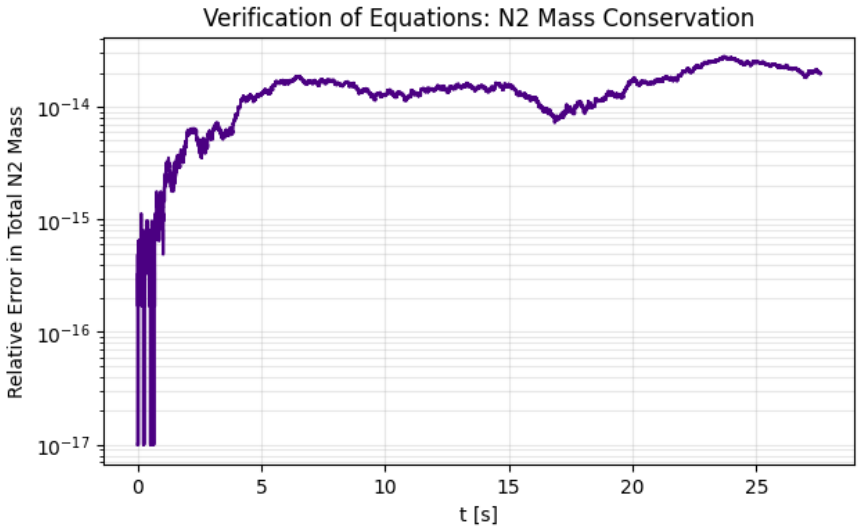


Figure 7: Relative error of total nitrogen mass ($m_p + m_u$) throughout the nominal burn.

Figure 7 demonstrates that our model conserves total nitrogen mass to near machine precision (relative error $< 10^{-15}$) throughout the entire transient burn. Furthermore, as an independent check of the integration loop, we overlaid our custom RK4 output with SciPy's established `solve_ivp` function utilizing the RK45 method.

As shown in Figure 8, the fuel mass flow trajectories are indistinguishable. The absolute discrepancy between the custom script and SciPy never exceeds 10^{-13} kg/s, proving our custom method robustly solves the physical equations as intended.

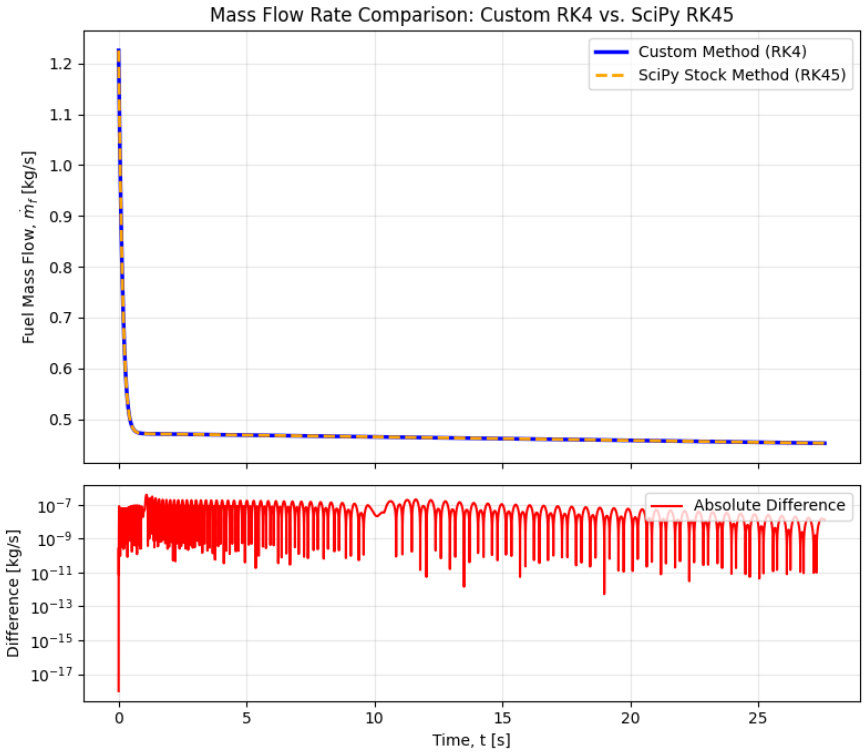


Figure 8: Comparison of fuel mass flow trajectories between the custom RK4 integrator and SciPy's RK45 implementation, including the absolute discrepancy.

3.2 Parameter Evaluation

Selecting an appropriate time step (Δt) for our simulation requires balancing absolute stability constraints against our engineering accuracy targets.

Our system exhibits stiffness due to the fast start-up transient of the ullage pressure dynamics. As established in the method section, the largest eigenvalue peaks at $|\lambda|_{max} = 8.53\ \text{s}^{-1}$ early in the burn. The absolute stability region for the RK4 method intersects the negative real axis at $w^* = 2.785$. Therefore, to prevent numerical blow-up, our time step is strictly bounded by stability at:

$$\Delta t_{stab} = \frac{w^*}{|\lambda|_{max}} = \frac{2.785}{8.53} = 0.326\ \text{s} \tag{47}$$

However, for this system to serve as an effective design tool, we imposed a strict accuracy requirement: the maximum relative error in the fuel mass flow rate ($\dot{m}_f$) must remain under 1%. Using a bisection search over our error sweep data, we determined that satisfying this 1% error tolerance requires a time step of:

$$\Delta t_{1\%} = 0.133\ \text{s} \tag{48}$$

Because $\Delta t_{1\%} < \Delta t_{stab}$, we conclude that *accuracy, rather than stability, dictates the time step parameter* for this IVP. Operating at $\Delta t = 0.133$ s provides a comfortable safety factor of ≈ 2.5 against the stability limit. Consequently, we utilize a fixed time step of $\Delta t = 0.1$ s for all subsequent operational results and parameter sweeps, guaranteeing both high-fidelity accuracy and unconditional stability over the duration of the burn.

4 Results

The initial regulator setpoint is 360 psig. Immediately after the burn starts, the ullage pressure drops significantly to ∼309 psig. After the initial drop, the ullage pressure remains constant at 309 psig, with only a slight pressure drop at the end of the burn. During the entire time of the burn, the engine chamber pressure remains constant at 300 psig. The pressure differential is always positive between the regulator and engine. This ensures that the flow will always be directed in the intended direction (Figure 9).

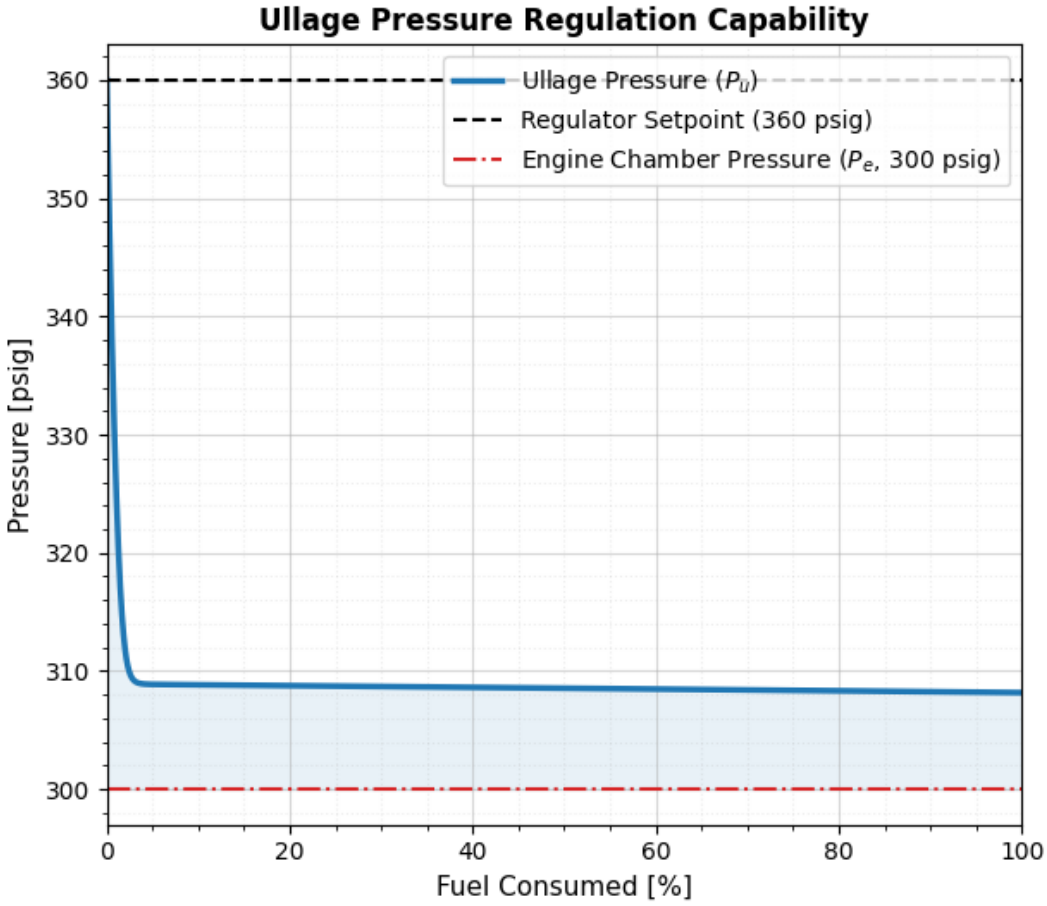


Figure 9: Ullage pressure regulation capability. The ullage pressure P_u drops rapidly from the 360 psig regulator setpoint to approximately 309 psig, then remains above the engine chamber pressure P_e (300 psig) for the entire burn.

Similar to the response of the ullage pressure to the burn, the fuel mass flow rate decreases rapidly at $t = 0$. The initial flow rate is ∼1.23 kg/s, but quickly settles to a moderately stable rate of 0.47 kg/s. As the burn fires, the fuel flow rate decreases slowly from 0.47 kg/s to 0.45 kg/s. Unlike the ullage pressure or fuel mass flow rate, the nitrogen pressurant tank pressure decreases linearly throughout the entire duration of the burn. The initial tank pressure is 3900 psig and decreases linearly with a final pressure of 3150 psig. Even with the decreasing tank pressure, the regulator supplies pressurant throughout the entire burn at a constant rate. This is exactly what the system was designed to accomplish. Furthermore, the fuel mass flow rate is nearly constant. This is a target of the system, and the graph verifies this requirement (Figure 10).

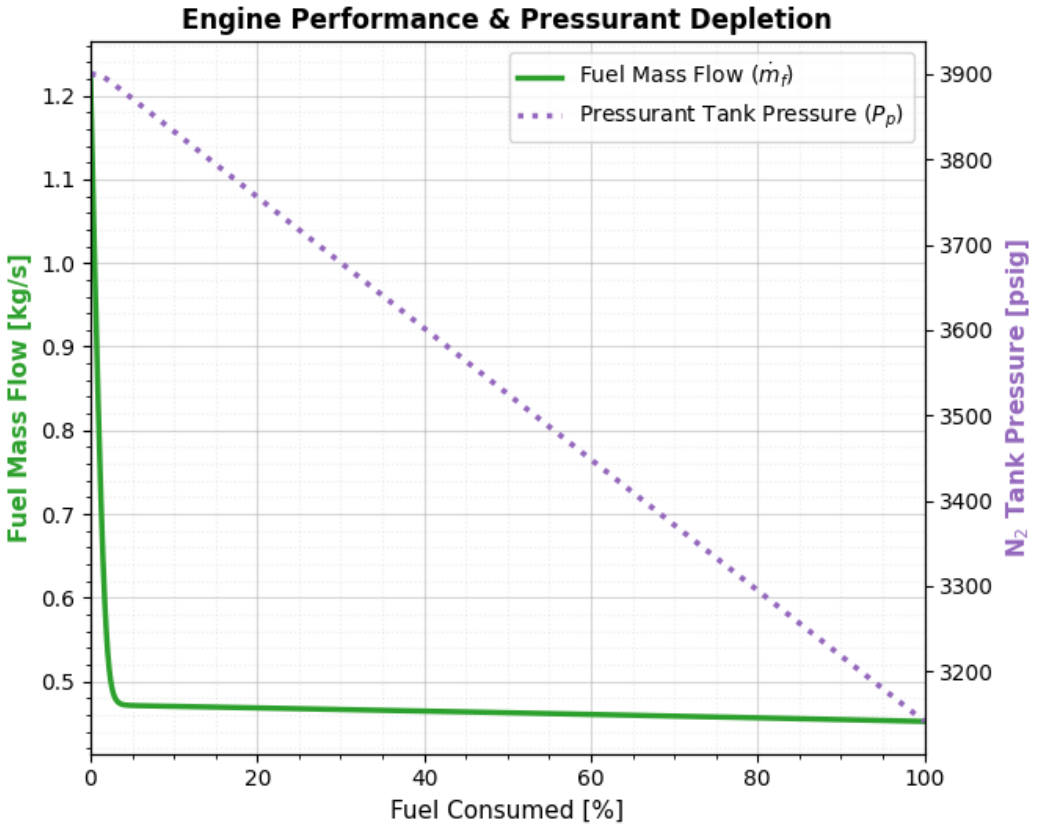


Figure 10: Engine performance and pressurant depletion. The fuel mass flow rate $\dot{m}_f$ (left axis) settles quickly to a nearly constant value, while the nitrogen pressurant tank pressure P_p (right axis) decreases linearly over the burn.

As demonstrated in the previous section, the custom method, as compared to the stock method, for integration, results in nearly identical results. The maximum difference between the two methods is on the order of 10^{-7} kg/s.

The custom RK4 method achieves an accuracy within the 1% error requirement. It accomplishes this with 207 steps and in 0.017 s. This was less than the other methods tested and still calculated the state accurately, as shown in the comparisons. Although the AB4 method used fewer function evaluations, its accuracy requirement places it close to its stability limit for this case. The custom RK4 method provides a balance between accuracy and computational cost for this system. Its performance makes it suitable for repeated simulations when evaluating alternative system configurations.

5 Contributions

The following section discusses each member contribution to the project. The work is distributed during in-person meetings, and is updated in text group chat. The estimated percentage of the member contribution is used to better visualize how the group project is divided.

Alex Knezevic $\approx 25\%$ Alex is responsible for the premise and the math for modeling the system so that it can be analyzed. They defined the problem and constraints, and converted the model into a form that we see in class and can apply methods to. They also wrote the premise in the report and the code for the model.

Aidan O'Sullivan $\approx 20\%$ Aidan is responsible for the results section of the report. He analyzed the response of the system to the physical meaning of the state variables. He wrote all the analysis, and used an LLM to translate the plain text into cleanly formatted LaTeXcode.

Arthur Kot $\approx 25\%$ Arthur Kot is responsible for the Correct Implementation section of the report. He analyzed the error for the report's implementation of RK4 on its equations, and compared the error to both the physical property of mass conservation between the two tanks present within the system and to SciPy's stock RK45 integrator. An LLM was then used to convert these analyses and graphs into LaTeX format for the report, and edited for proper flow through the text.

Athiwat Chotrapeepirom $\approx 30\%$ Athiwat is responsible for the method section, which corresponds to rubric item (2). He analyzes different numerical methods with the consideration of accuracy, stability, and stiffness. He then puts all these factors into cost considerations. This means he ultimately chooses the numerical method, which is RK4, for this project. The entire method section works including the code work, plots, and derivation write-up are done by him with the aid of Large Language Model. The use of LLM is discussed in the following Acknowledgments section.

6 Acknowledgments

Alex Knezevic Alex used Claude for creating the two diagram figures, taking the core relations and equations into a state space model, and a majority of the code implementing the modeled system to usable functions. All writing and system definition is untouched by any LLM.

Aidan O'Sullivan I used Claude Sonnet 5.5 for the translation of my work into LaTeX. The LLM did not change any of my words or ideas, but instead formatted the words into code.

Arthur Kot Google Gemini 3.1 Pro was used to format thoughts into LaTeX and debug code. The method of error analysis and comparison against a real world law was explained to the LLM during this process to minimize errors during debugging and translation.

Athiwat Chotrapeepirom I use Claude Opus 5.5 to help derive the method section. My workflow started with my descriptive approach to the method section, explaining to the model how I wanted to test the accuracy, stability, and stiffness of the IVP. Then it gave me suggestions for my approach, what derivation I need to do, and organized my argument order. After looking at its response, I further tailored it to approach the IVP following the in-class lectures. The majority of the code is generated using Claude, and I have checked and confirmed its correctness to my liking. Additionally, I used Claude to help explain some confusing concepts throughout the section.